

Optimization in Data Analysis

Project 5.3: Minimization of Weakly Convex Functions

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June 2026

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Introduction to Weakly Convex Functions

Definition

A function $\phi : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is called ρ -weakly convex ($\rho > 0$) if the assignment

$$x \mapsto \phi(x) + \frac{\rho}{2}\|x\|^2$$

is convex.

- Bridges the gap between standard convex optimization and chaotic non-convex problems.
- Features a quadratic under-estimator instead of a linear one:

$$\phi(y) \geq \phi(x) + \langle v, y - x \rangle - \frac{\rho}{2}\|y - x\|^2 \quad \forall v \in \partial\phi(x)$$

Smoothing via the Moreau Envelope

Nonsmooth optimization lacks smooth gradients. We analyze convergence using the **Moreau Envelope**:

$$\phi_\lambda(x) = \inf_y \left\{ \phi(y) + \frac{1}{2\lambda} \|y - x\|^2 \right\}$$

Key Structural Properties (for $\lambda < \rho^{-1}$):

- ϕ_λ is continuously differentiable (C^1 -smooth).
- $\nabla \phi_\lambda(x) = \frac{1}{\lambda}(x - \text{prox}_{\lambda\phi}(x))$.
- The gradient norm $\|\nabla \phi_\lambda(x)\|$ serves as a metric for stationarity.

Optimization Goal

Find an ϵ -stationary target point where:

$$\|\nabla \phi_\lambda(x)\| \leq \epsilon$$

Methodology: Model-Based Stochastic Framework

We want to minimize stochastic models built from $\phi(x)$.

Stochastic Model-Based Step

For each sample ξ_t , the update rule follows:

$$x_{t+1} = \arg \min_x \left\{ r(x) + f_{x_t}(x, \xi_t) + \frac{\beta_t}{2} \|x - x_t\|^2 \right\}$$

where $f_{x_t}(x, \xi_t)$ is a model of the loss around the point x_t , and β_t is a regularizing proximal weight.

- **Stochastic Subgradient Method:** Regularizes a linear mapping constructed via an i.i.d subgradient sample.
- **Proximal-Linear Method:** Exploits composite functions $\phi(x) = h(c(x))$ by explicitly linearizing the inner expression $c(x)$.
- **Stochastic Proximal Point Method:** Uses the exact true objective function as the model with no approximation at all.

Reproduction: Phase Retrieval ($d = 10$)

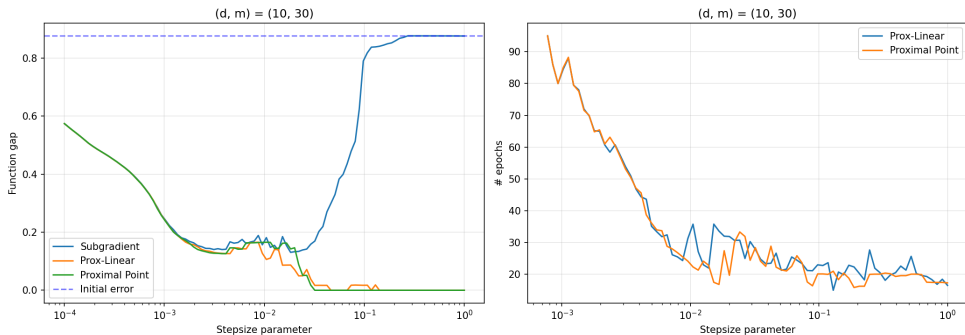


Figure 1: Phase retrieval reproduction: $(d, m) = (10, 30)$

Problem Formulation & Setup: Recover a vector $x \in \mathbb{R}^d$ from magnitude-only observations $b_i = |\langle a_i, x \rangle|$. Initial parameters are set to $d = 10$ (dimension) and $m = 30$ (sample size).

Phase Retrieval: Intermediate Dimension ($d = 50$)

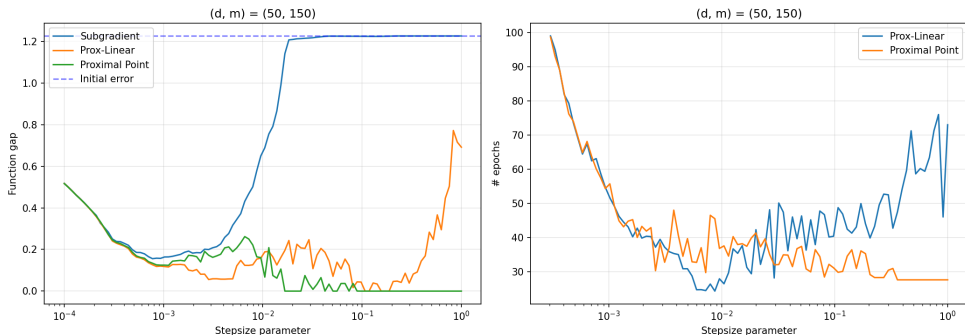


Figure 2: Phase retrieval reproduction: $(d, m) = (50, 150)$

Setup Expansion: The test space scales to dimension $d = 50$ and sample size $m = 150$. The proximal point method continues to exhibit robust and consistent convergence.

Phase Retrieval: Higher Dimension ($d = 100$)

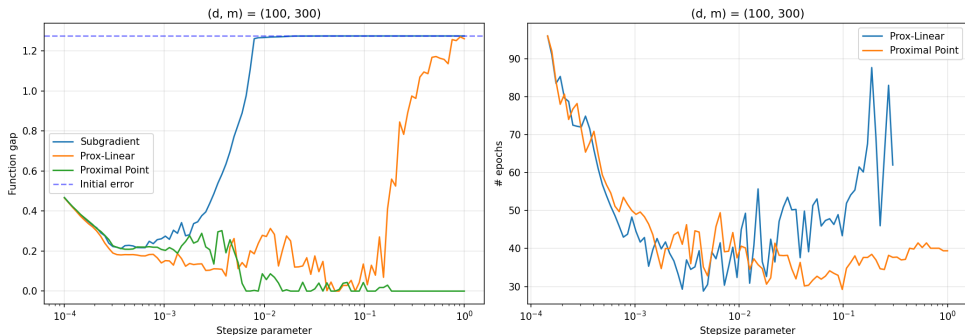


Figure 3: Phase retrieval reproduction: $(d, m) = (100, 300)$

Setup Expansion: The test space scales to dimension $d = 100$ and sample size $m = 300$. The proximal point method continues to exhibit robust and consistent convergence.

Reproduction: Blind Deconvolution ($d = 10$)

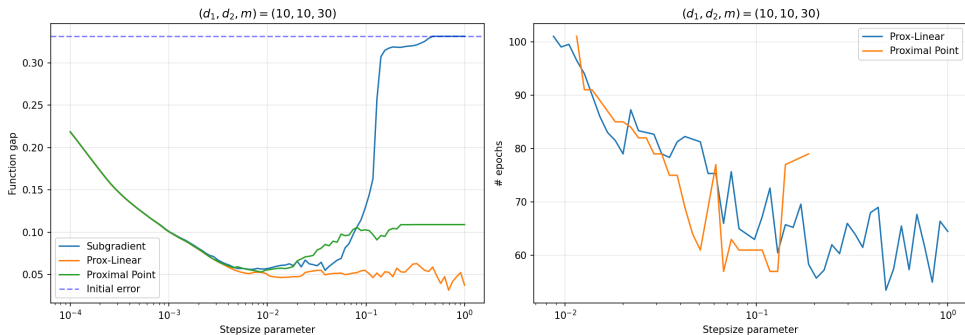


Figure 4: Blind deconvolution reproduction: $(d_1, d_2, m) = (10, 10, 30)$

Problem Formulation & Setup: Simultaneously recover bilinear signals $u \in \mathbb{R}^{d_1}$ and $v \in \mathbb{R}^{d_2}$ from data pairs $y_i = u^T A_i v$. The initial baseline validation uses parameters $d_1 = d_2 = 10$ and a sample size of $m = 30$.

Blind Deconvolution: Medium Scale ($d = 50$)

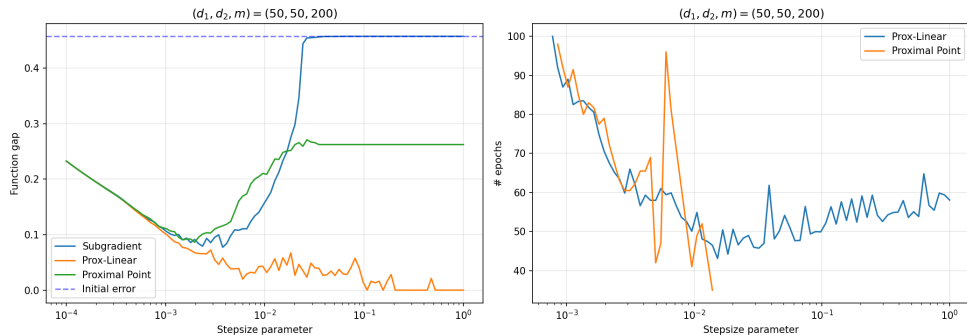


Figure 5: Blind deconvolution reproduction: $(d_1, d_2, m) = (50, 50, 200)$

Setup Expansion: Parameters scale up to dimensions $d_1 = d_2 = 50$ with a sample size of $m = 200$. The functional gap convergence profiles indicate a stability advantage for the proximal point model.

Blind Deconvolution: High Dimension ($d = 100$)

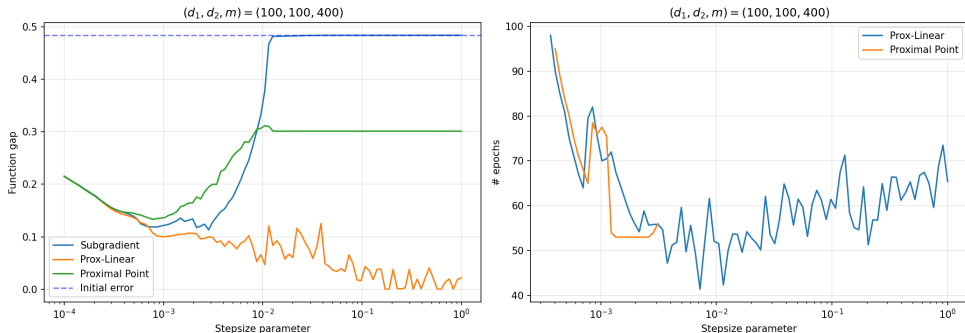


Figure 6: Blind deconvolution reproduction: $(d_1, d_2, m) = (100, 100, 400)$

Setup Expansion: The final experiment scales the problem to dimensions $d_1 = d_2 = 100$ and sample size $m = 400$.

Comparison with Gradient-Based Optimizers ($d = 10$)

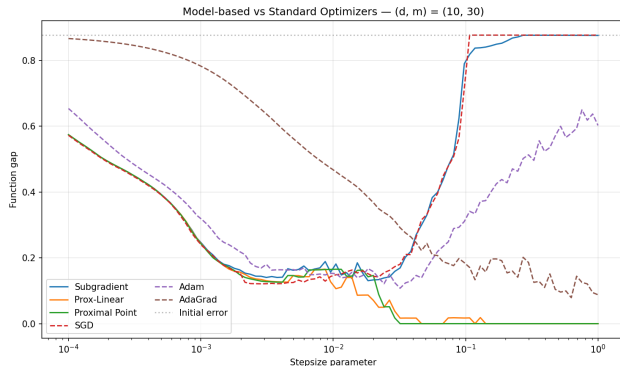


Figure 7: Optimizer comparison: $(d, m) = (10, 30)$

Key Insights: The model variants are benchmarked directly against mainstream first-order choices: Adam, AdaGrad, and standard SGD. Dedicated weakly convex model-based methods show steadier optimization paths without suffering from changing structural oscillations.

Optimizer Comparison: Medium Scale ($d = 50$)

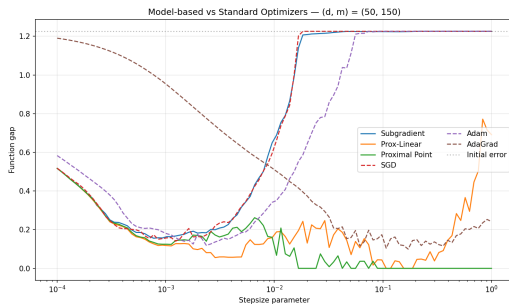


Figure 8: Optimizer comparison: $(d, m) = (50, 150)$

Observations: Regular generic learning optimizers (like SGD) and the subgradient method fall apart and fail to converge when the step-size is too large. Conversely, the model-based prox-linear and proximal-point routines successfully reduce the global function gap over a much wider, forgiving range of step-sizes.

Optimizer Comparison: High Dimension ($d = 100$)

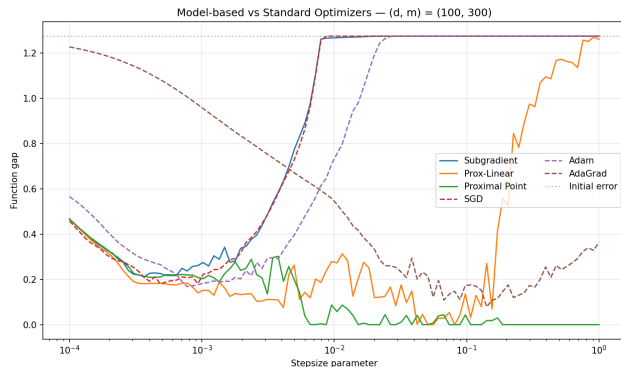


Figure 9: Optimizer comparison: $(d, m) = (100, 300)$

Structural Scaling: Adaptive optimizers like Adam fail to exploit the underlying weakly convex structure, leading to worse performance compared to proximal model-based methods.

Moreau Envelope Gradient Norm ($d = 10$)

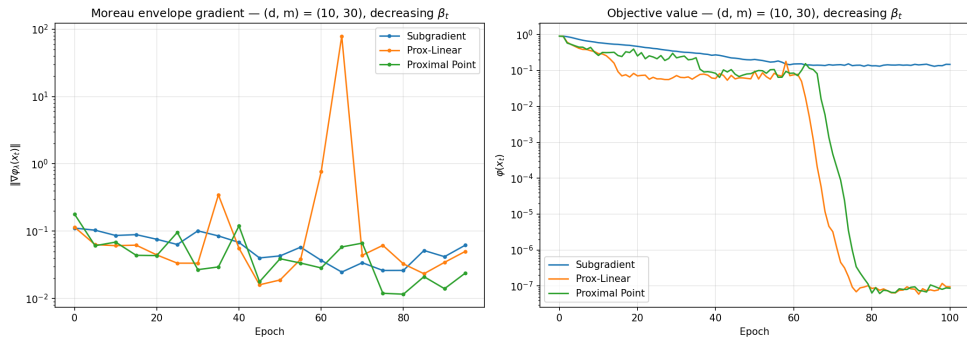


Figure 10: Moreau norm tracking: $(d, m) = (10, 30)$

Stationarity Metric: Evaluation of both the smoothed Moreau gradient profile and the companion objective function values across epochs. This metric explicitly confirms robust numerical verification of asymptotic properties in small-scale limits.

Moreau Envelope Gradient Norm ($d = 50$)

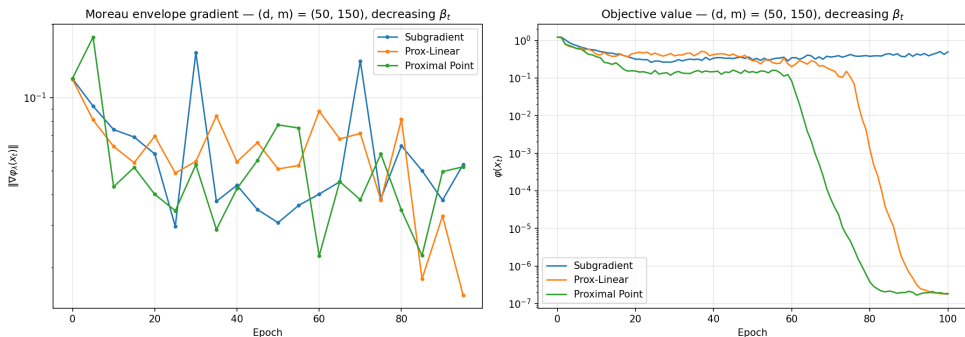


Figure 11: Moreau norm tracking: $(d, m) = (50, 150)$

Stationarity Metric: Continuous monitoring of the metric $\|\nabla\phi_\lambda(x)\|$ shows an overview of performance, offering concrete numerical confirmation of stable progress directly toward true subgradient stationarity. The proximal point model decreases the fastest.

Moreau Envelope Gradient Norm ($d = 100$)

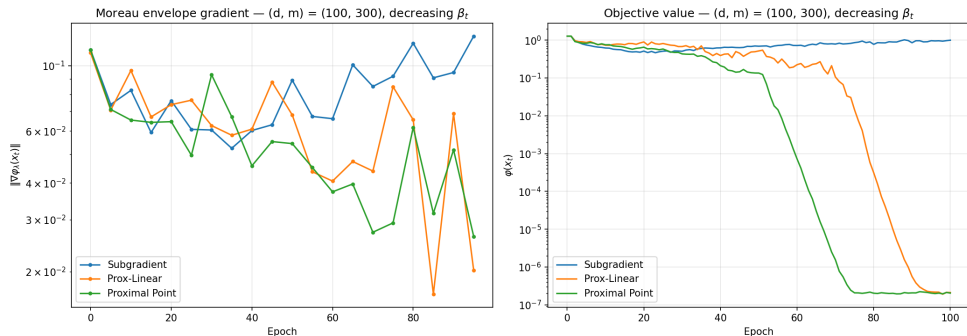


Figure 12: Moreau norm tracking: $(d, m) = (100, 300)$

High-Dimensional Smoothing: Unlike the highly erratic raw objective function values, which suffer from local nonsmooth fluctuations, tracking the Moreau gradient norm effectively smooths out subgradient shifts, revealing true global convergence behavior.

Evaluating Dynamic Stopping Criteria

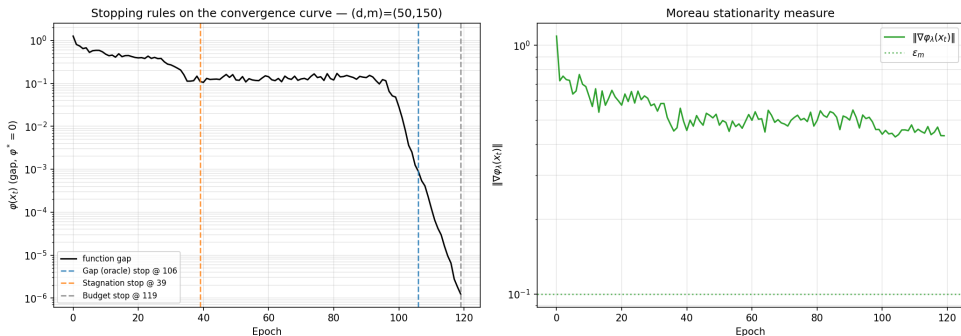


Figure 13: Four stopping rules evaluation: $(d, m) = (50, 150)$

Tested Stopping Rules: Moreau norm, oracle gap, iterate stagnation, and time budget.

Critical Findings: Iterate stagnation stopped prematurely on flat plateaus, and the Moreau norm unexpectedly plateaued near 0.7 instead of dropping.

Empirical Verification of $O(k^{-1/4})$ Complexity

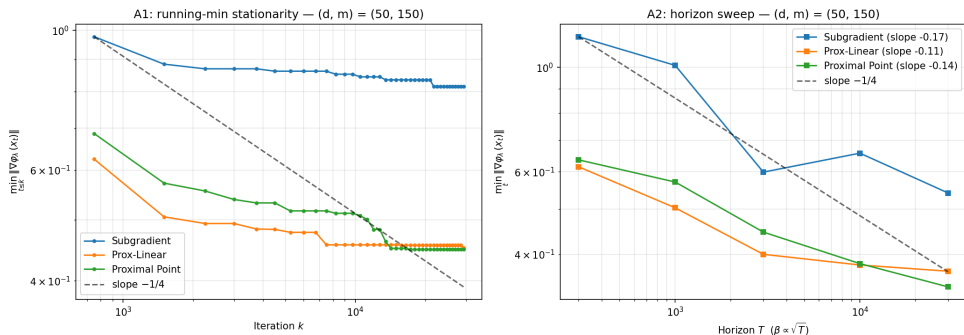


Figure 14: Empirical evaluation of the $O(k^{-1/4})$ convergence rate with $(d, m) = (50, 150)$.

- **Left panel (A1):** Running-minimum stationarity track across a single long deployment.
- **Right panel (A2):** Achieved stationarity vs. horizon T , step parameter scaling $\beta \propto \sqrt{T}$.
- Experimental slopes are slightly shallower than the theoretical worst-case bound of $-\frac{1}{4}$.

Empirical Recovery Probability (Phase Transition)

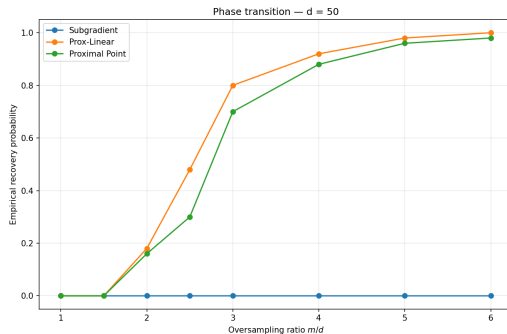


Figure 15: Recovery probability vs. oversampling ratio m/d .

Recovery Threshold: Statistical metrics are aggregated from 50 independent empirical trials per data ratio point with baseline dimension $d = 50$. Near-perfect global recovery occurs systematically when the ratio reaches $m/d \geq 5$.

Conclusions

- **Theoretical Validation:** Experimental findings align with the fundamental $O(k^{-1/4})$ convergence behavior, although empirically fitted slopes were shallower due to rapid convergence to the noise floor and low initial errors.
- **Algorithmic Structural Gains:** Custom model-based algorithms (such as the Proximal-Linear variant) substantially lead over generic routines (Adam, SGD) in weak-convex landscapes.
- **Phase Transitions:** Effective signal retrieval features precise structural sampling transition targets ($m/d \geq 5$).

Thank you!